

END TERM EXAMINATION

FIFTH SEMESTER [BCA] DECEMBER 2024

Paper Code: BCA301

Subject: Operating System & Linux Programming
(Batch 2021 Onwards)

Time: 03:00 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.No.1 which is compulsory.
Select one question from each unit

(2.5×10=25)

Q1 Answer the following:

- Explain process states with the help of a diagram.
- Explain thrashing.
- Write Linux commands to display currently running processes, terminate a process, execute a command in background, display all jobs created, bring a job in foreground.
- Differentiate between I/O bound and CPU bound processes.
- Explain the working modes of vi editor.
- What is PCB? State the contents of PCB.
- Distinguish between internal and external fragmentation.
- What is Starvation and Aging?
- What is Belady's Anomaly?
- What is Swapping in OS?

UNIT-I

- Q2 a) What is an Operating System? State the functions of an Operating System. (6.5)
b) Define the essential properties of the following types of Operating Systems: (6)

i. Batch ii. Time Sharing iii. Real Time

OR

- Q3 a) With the help of diagram, explain the architecture of LINUX operating system. (7.5)
b) State and explain the Basic Directory Structure of Linux OS. (5)

UNIT-II

- Q4 a) What are the criteria for CPU scheduling? (5)
b) Consider following processes with length of CPU burst time in milliseconds (7.5)

Process	Burst time	Arrival Time
P1	5	0
P2	10	2
P3	2	4
P4	1	7

All process arrived in order p1, p2, p3, p4.

- Draw Gantt charts illustrating execution of these processes for FCFS, preemptive SJF and round robin (quantum=3).
- Calculate waiting time for each process for each scheduling algorithm.
- Calculate average waiting time for each scheduling algorithm.

OR

P.T.O.

- Q5 a) Explain the terms relating to IPC – a) Critical Section b) Race Condition c) Mutual Exclusion (6.5)
- b) Explain Semaphores and its types. (6)

UNIT-III

- Q6 a) What is deadlock? List and explain conditions for the deadlock situation and various ways for deadlock prevention. (7.5)
- b) Consider the following snapshot of a system: (5)

Threads	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
T_0	6	0	1	2	4	0	0	1	3	2	1	1
T_1	2	7	5	0	1	1	0	0				
T_2	2	3	5	6	1	2	5	4				
T_3	1	6	5	3	0	6	3	3				
T_4	1	6	5	6	0	2	1	2				

Answer the following questions using the Banker's algorithm:

- a. Find the total number of resources of type A, B, C, D.
- b. What is the content of the matrix Need?
- c. Is the system in a safe state? If it is, find the safe sequence.

OR

- Q7 a) Explain the concept of Contiguous memory allocation. Consider free memory holes of sizes 40K, 5K, 28K, 15K, 27K are available. The processes of size 29K, 12K, 25K, 28K is to be allocated. How are the processes placed in first fit, best fit, worst fit. Which algorithm makes the most efficient use of memory? (6)
- b) Explain the concept of demand paging. Consider the following page reference string, and assume 4-page frames (6.5)

7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 1, 0, 5, 4, 6, 2, 3, 0, 1

Explain the following page replacement algorithms of demand paging and compute the number of page faults for each algorithm.

- i. LRU replacement ii. FIFO replacement iii. Optimal replacement

UNIT-IV

- Q8 a) Explain any three schemes for defining the logical structure of a directory (6.5) also state the operations that could be performed on directories.
- b) Compute the total head movements according to FCFS, SCAN and C-SCAN HDD Scheduling (6)
- Queue = 68, 183, 67, 122, 14, 124, 95, 37
- Head starts at 35

OR

- Q9 a) Current permissions of file **demo.txt** are "rw-r-xr--". Using both relative and absolute mode, specify the 'chmod' expression required to change the permissions as follows (6)
- i. **rxwxrwxrwx** ii. **r-- r-- ---** iii. **--- --- ---** iv. **rxwxr-xr-x**
- b) Explain the various types of files in Linux and the file Permissions applied to protect the file. (6.5)

END TERM EXAMINATION

FIFTH SEMESTER (BCA) DECEMBER-2024 JANUARY-2025

Subject: Cloud Computing

Paper Code: BCA-305

Maximum Marks: 75

Time: 3 Hours

Note: Attempt any five questions in all including Q.No. 1 which is compulsory. Select one question from each unit.

(2.5x10=25)

Q1 Attempt all Questions:

- (a) What is Cloud Computing?
- (b) Difference between elasticity and scalability in cloud computing?
- (c) Explain Service Oriented Architecture (SOA)?
- (d) What is Secure Sockets Layer (SSL)?
- (e) Explain Hybrid Cloud?
- (f) Discuss the principles of security in cloud computing?
- (g) Define Digital signature with example?
- (h) Elaborate Cloud computing economics.
- (i) What is software productivity in cloud?
- (j) Define virtualization with example?

UNIT-I

Q2 Explain the vision of cloud computing which includes- Characteristics, benefits and Challenges? (12.5)

OR

Q3 (a) Draw the diagram of cloud computing and explain its diagram? (6.5)
(b) Explain services of Internet? (6)

UNIT-II

Q4 (a) What is Cloud reference model? Explain with example? (6.5)
(b) What are computing platform and technologies? (6)

OR

Q5 (a) What are the advantages of "Platform as a Service" (PaaS)? (6.5)
(b) What are pros and cons in comparison of Public vs. Hybrid cloud? (6)

UNIT-III

Q6 (a) Differentiate between Distributed Vs Parallel computing? (6.5)
(b) What are the approaches of parallel programming? (6)

OR

Q7 (a) What are the Elements of parallel computing? (6.5)
(b) What are the principles of Computing? (6)

UNIT-IV

Q8 (a) What are the characteristics of visualized environments? (6.5)
(b) Discuss Pros and cons of virtualization? (6)

OR

Q9 (a) What are the benefits of virtualization using in Cloud Computing? (6.5)
(b) Differentiate between Full virtualization and Partial virtualization? (6)

(6)
(6.5)

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END TERM EXAMINATION

FIFTH SEMESTER [BCA] DECEMBER 2024

Paper Code: BCA311

Subject: Machine Learning with Python

Time: 03:00 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit

Q1 Write short notes on the following (Any Five)

(5×5=25)

- a) What do you mean by R2 score?
- b) What are some real-life applications of clustering algorithms?
- c) Why do we take smaller values of the learning rate?
- d) What is bagging?
- e) Do we require feature scaling for Decision Tree Classifier? Explain
- f) How to identify whether the model has over fitted the training data or not?
- g) What do you mean by perceptron?
- h) List down the names of applications of Artificial Neural Networks.

Q2 What is logistic regression and how does it differ from linear regression? (12.5)

OR

- Q3 a) What are the various types of machine learning? Explain. (6)
b) What are the various Classification metrics? Explain in detail. (6.5)

- Q4 a) Given entropy of parent = 1, weights averages = (3/4, 1/4) and entropy of children = (0.9, 0). What is the information gain? (4.5)
b) Explain KNN algorithm. Perform the KNN Classification on following dataset and predict the class of X[P1=3, P2=7], where k=3. (8)

P1	P2	Class
7	7	False
7	4	False
3	4	True
1	4	True

OR

- Q5 a) Explain Support Vector Machine. Define the terms Hyperplane, Support Vectors, Kernel, Hard and Soft Margin. (6)
b) What are the ensemble learning models? Explain Random Forest in detail. (6.5)

- Q6 a) How does over fitting can be avoided using regularization? (5)
b) What is HebbNet in machine learning? Explain Hebbian Learning Rule. (7.5)

OR

- Q7 a) What is the role of the Activation functions in Neural Networks? List down the names of some popular Activation Functions used in Neural Networks. (6)
b) What do you mean by back propagation? Explain its working. (6.5)

P.T.O.

- Q8 a) What do you mean by clustering? How do you decide on the number of clusters(k) in a K-Means algorithm? (8.5)
b) Explain the concept of "Curse of Dimensionality". (4)

OR

- Q9 a) What do you mean by Principle Component Analysis? How it helps in feature extraction? (6.5)
b) What is the Agglomerative Hierarchical Clustering? Explain the different linkage methods used in the Hierarchical Clustering Algorithm. (6)

END TERM EXAMINATION

FIFTH SEMESTER [BCA] DECEMBER 2024

Paper Code: BCA-303

Subject: Computer Graphics
(BATCH 2021 ONWARDS)

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.No 1 which is compulsory. Select one question from unit.

- Q1 Attempt **any five** of the following questions: (5x5=25)
- a) Differentiate between random scan and raster scan display devices.
 - b) Explain the working of a video controller.
 - c) Define the term si) Polygon mesh ii) Aspect ratio iii) Persistence.
 - d) List three properties of a Hermite curve. Explain Convex hull of a Bezier curve.
 - e) Write down the matrix equations of 3D rotation & scaling transformations.
 - f) Discuss the disadvantages of using Digital Differential Analyzer algorithm for scan converting lines.
 - g) Distinguish between orthographic and oblique projections.
 - h) What do you understand by parametric continuity? How is it different from geometric continuity, explain?

Unit-I

- Q2 a) What is eight-point symmetry of a circle? Explain Bresenham's line drawing algorithm with its derivation. (7)
- b) Explain the different application areas of computer graphics. (5.5)

OR

- Q3 a) Write notes on – (6)
- I. Vertical & Horizontal Retrace
 - II. Disadvantages of DDA line drawing algorithm
 - III. Aliasing v/s antialiasing
- b) Consider three different raster systems with resolutions of 640 x 480, 1280 x 1024, and 2560 x 2048. What size is frame buffer (in bytes) for each of these systems to store 24 bits per pixel? (6.5)

Unit-II

- Q4 a) Prove that two successive 2-D translations are additive in nature, i.e. $T(T_{x1}, T_{y1}).T(T_{x2}, T_{y2}) = T(T_{x1}+T_{x2}, T_{y1}+T_{y2})$ (6)
- b) Explain window to viewport transformation with derivation. (6.5)

OR

- Q5 a) Define clipping. Explain Cohen Sutherland line clipping algorithm with the help of a suitable diagram. (6)
- b) What is 2-D transformation? Perform a 45° counter clockwise rotation of a triangle with vertices A(0,0), B(1,1), C(5,2) about the origin. Find the new coordinates of the vertices of the triangle. (6.5)

P-1/2
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P.T.O.

Unit-III

- Q6 Consider a Bezier curve having control points $P_1(20,0)$, $P_2(0,20)$, $P_3(80,40)$ and $P_4(40,0)$.
Find the coordinates of the curve when $t=\{0.0, 0.2, 0.6, 1.0\}$.
Draw a rough sketch of the curve and the convex hull. (12.5)

OR

- Q7 a) Explain different surface rendering techniques used in Computer Graphics. (8)
b) Discuss Phong model of reflection. List different components of the model. (4.5)

Unit-IV

- Q8 a) What do you understand by hidden surface removal? Explain painter's algorithm for hidden surface removal. (7)
b) Explain various types of parallel projections. (5.5)

OR

- Q9 Explain the different types of projections with proper sketches. (12.5)

P-2/2
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